

Geometry: Unit 12, Lesson 9

1. The standard form of a circle is shown $x^2 + y^2 + 6x - 12y - 17 = 0$. Some of the steps Jenelda took to write the equation of the circle in center-radius form are shown.

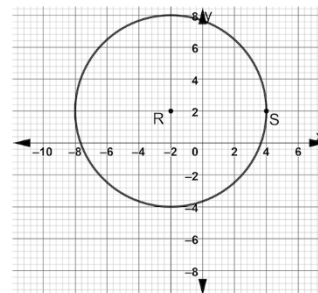
Step	Jenelda's Work
1.1.	$x^2 + y^2 + 6x - 12y - 17 = 0$
2.2.	$x^2 + 6x + \quad y^2 - 12y = 17$

Some of Jenelda's third step is shown. $x^2 + 6x + \boxed{} + y^2 - 12y + \boxed{} = 17 + \boxed{} + \boxed{}$

Some of Jenelda's fourth step is shown. $(x + \boxed{})^2 + (y - \boxed{})^2 = 17 + \boxed{} + \boxed{}$

What value should Jenelda put in the red, blue, orange, and purple boxes?

2. Circle R is shown on the coordinate plane. What are the coordinates of the center of Circle R ? What is the radius of circle R ?



3. In a circle whose equation is $(x + 7)^2 + y^2 = 64$. What are the coordinates of its center? What is the circles radius?

